Differentiation: Introduction

Questions 1–17

Question 1. Find $\dfrac{dy}{dx}$ for each of these:

a) $y = 4x^3 + 5x^2$
b) $y = 2x - 9x^2$
c) $y = x^6 - 18x^2 + 3$
d) $y = x^{-1} + x^4$
e) $y = x^3 - x^{-2}$
f) $y = x^2 + x^{-2} + x + 1$
g) $y = 3x^5 - 2x^{-2} - x + 6$
h) $y = 2x^a - x^b$

Question 2. Find $f'(t)$ for each of these:

a) $f(t) = \dfrac{1}{2}t^2 + \dfrac{3}{5}t^{-5}$
b) $f(t) = \dfrac{1}{4}t - \dfrac{3}{4}t^4$
c) $f(t) = \dfrac{1}{12}t^6 + 10t^2 + 7$
d) $f(t) = \dfrac{1}{3}t^{-1} + \dfrac{2}{3}t^6$
e) $f(t) = \dfrac{7}{8}t^2 - \dfrac{1}{4}t^{-3} - 4$
f) $f(t) = \dfrac{2}{7}t^8 + \dfrac{1}{7}t^{-3} + \dfrac{1}{5}t + \dfrac{4}{7}$
g) $f(t) = -\dfrac{1}{4}t^6 - \dfrac{1}{2}t^{-2} + 2t + \dfrac{5}{9}$
h) $f(t) = -\dfrac{1}{2}t^{a/b}$, $b \neq 0$

Question 3. Find $\frac{dy}{dt}$ for each of these:

a) $y = 8t^{1/2} + 6t^{1/3}$
b) $y = 5t - 12t^{1/4}$
c) $y = t^{2/3} + 7t^{3/5}$
d) $y = t^{-1/2} + t^{1/4} + \frac{1}{4}$
e) $y = -t^{5/8} - t^{-3/8} - \frac{1}{8}t + \frac{1}{8}$
f) $y = t^{1/3} + t^{-1/3} + 3t + 3$
g) $y = -t^{3/4} + t^{-8/9} - \frac{1}{12}t + 1$

Question 4. Find $f'(x)$ for each of these:

a) $f(x) = x(x + 3)$
b) $f(x) = (x + 1)(x + 5)$
c) $f(x) = (x - 2)(2x + 5)$
d) $f(x) = (x + 4)^2$
e) $f(x) = x^2(x - 1)$
f) $f(x) = (x + 2)^2(5x - 1)$
g) $f(x) = x(7x - x^3)$
h) $f(x) = x^2(9x - x^{-1})$
i) $f(x) = x(7x^{-1} - x^{-2} + 8)$
j) $f(x) = \left(x + \frac{1}{x}\right)^3$
k) $f(x) = (x + p)(x + q)$
l) $f(x) = (x - a)\left(x + \frac{1}{a}\right)$, $a \neq 0$

Question 5. Write each of the following in the form x^k:

a) $\sqrt{x}$
b) $\sqrt{x^5}$
c) $\dfrac{1}{x^4}$
d) $(x\sqrt{x})^3$
e) $\sqrt[3]{x}$
f) $\dfrac{1}{\sqrt{x}}$
g) $\dfrac{x}{\sqrt{x}}$
h) $\dfrac{1}{x\sqrt{x}}$
i) $\dfrac{1}{x \times \sqrt[a]{x}}$ where $a > 0$

Question 6. Write each of the following in the form ax^b:

a) $5\sqrt[3]{x}$
b) $\sqrt{4x^3}$
c) $\dfrac{3}{x^2}$
d) $(2x\sqrt{x})^5$
e) $\dfrac{\sqrt[8]{x}}{3}$
f) $\dfrac{4}{\sqrt{x}}$
g) $\dfrac{3x}{\sqrt{9x^4}}$
h) $\dfrac{5}{x\sqrt{16x}}$
i) $\dfrac{2}{\sqrt{25ax}}$ where $a > 0$

Question 7. Find $\frac{dy}{dx}$ for each of these:

a) $y = \sqrt[4]{x}$

b) $y = \sqrt{9x^5}$

c) $y = \dfrac{5}{x^3}$

d) $y = (x\sqrt{x})^3$

e) $y = \dfrac{\sqrt[3]{x}}{5}$

f) $y = \dfrac{3}{\sqrt{x}}$

g) $y = \dfrac{x}{\sqrt{x^4}}$

h) $y = \dfrac{2x}{\sqrt{16x^4}}$

Question 8. Find $f'(x)$ for each of these:

a) $f(x) = \sqrt{x}(x + 2)$

b) $f(x) = \sqrt{x}(x^2 - 1)$

c) $f(x) = \dfrac{x^3 + 4x}{x}$

d) $f(x) = \dfrac{2x^3 + 4x^{-2}}{x}$

e) $f(x) = \dfrac{9x - 1}{x^2}$

f) $f(x) = \dfrac{x^2 + 2x - 1}{\sqrt{x}}$

g) $f(x) = \dfrac{x^2 - 5x^{-3/2} + 1}{\sqrt{x}}$

Question 9. For each of the following, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$:

a) $y = x^3 + x^{-3} + 2x^2 - 5x + 8$

b) $y = \frac{1}{12}x^4 + \frac{1}{8}x^2 + 7x$

c) $y = -x^3 - 2x^{-1/2} + 15x$

d) $y = -x^{1/7} - x^{-1/7} - \frac{1}{9}x + \frac{1}{9}$

e) $y = x^3(2x + x^{-2})$

f) $y = \sqrt{16x^3}$

g) $y = \frac{2}{\sqrt{x}}$

h) $y = \frac{x^5 + 4x^{-2}}{x}$

Question 10. Find the value of the gradient of the curve at the given points:

a) $y = x^2 + 5x - 1$ at $x = 1$

b) $y = x^3 - 2x + 4$ at $x = -2$

c) $y = \frac{1}{\sqrt{x}} - 2x^2$ at $x = 4$

d) $y = \sqrt{x} - x$ at $(4, -2)$

e) $y = \frac{x^4 - 3x^3}{x}$ at $(2, -4)$

Question 11. Find $f''(1)$ for each of these:

a) $f(x) = x^3 - 5x^2 + 2$

b) $f(x) = \dfrac{1}{x^2} - x^2$

c) $f(x) = \sqrt{x} + 2x^2$

d) $f(x) = x\sqrt[3]{x} - 4x$

e) $f(x) = p\sqrt{x} - \dfrac{4}{qx}$ where $q \neq 0$

Question 12. Prerequisite knowledge for Q12–Q17: Differentiation of the form ae^{kx}

Find $f'(x)$ for each of these:

a) $f(x) = e^{2x}$

b) $f(x) = e^{13x}$

c) $f(x) = e^{-8x}$

d) $f(x) = e^{x/5}$

e) $f(x) = e^{-4x/7}$

f) $f(x) = 2e^{3x}$

g) $f(x) = 5e^{-x}$

h) $f(x) = \dfrac{1}{4}e^{-2x}$

i) $f(x) = \dfrac{1}{e^{5x}}$

j) $f(x) = \sqrt{e^{3x}}$

k) $f(x) = \dfrac{e^x}{\sqrt{e^x}}$

l) $f(x) = \dfrac{e^x}{\sqrt{e^{6x}}}$

m) $f(x) = \sqrt{e^{ax}}$

n) $f(x) = \dfrac{e^x}{\sqrt{e^{ax}}}$

Question 13. Find $\frac{dy}{dx}$ for each of these:

a) $y = 2e^{2x} - 5e^{3x}$
b) $y = 2x - e^{3x}$
c) $y = e^{-x} + x^5$
d) $y = x^5 - \frac{1}{4}x^4 + 3e^{-x/5}$
e) $y = \sqrt{x} + \frac{3}{e^{2x}}$
f) $y = \sqrt{ax} + \frac{5}{e^{bx}}$

Question 14. Find $f'(x)$ and $f''(x)$ for each of these:

a) $f(x) = e^{3x} + 8e^{-x}$
b) $f(x) = 5x^2 - e^{5x}$
c) $f(x) = 2e^{-x} + x^2$
d) $f(x) = x^4 - 8e^{-x/2} - \frac{1}{2}x^{-3}$

Question 15. Find the value of the gradient of the curve at the given points, giving your answers in exact form:

a) $y = e^x$ at $x = 2$

b) $y = 5e^x$ at $x = -1$

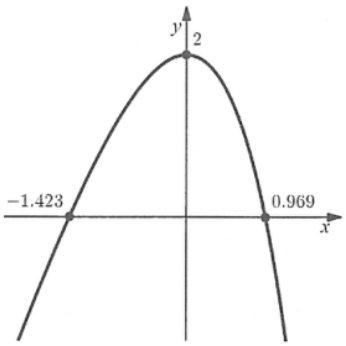
c) $y = x^2 - e^x$ at $(0, -1)$

d) $y = x - e^x$ at $(1, 1 - e)$

e) $y = 2(x + e^{2x})$ at $\left(\frac{1}{2}, 1 + 2e\right)$

f) $y = -3(x + e^{ax})$ at $(1, -3 - 3e^a)$

Question 16. The graph below shows the curve $y = 3(x - e^x) + 5$. Find the value of the gradient at each of the points where the curve meets the axes giving your answers to 2 decimal places.



Question 17. Find the values of k and p such that the second derivative of $y = \dfrac{k}{e^{px}}$ is always positive.